RVS COLLEGE OF ARTS AND SCIENCE(AUTONOMOUS), SULUR

B.Sc END SEMESTER PRACTICAL EXAMINATIONS NOV-2025

B.Sc DATA SCIENCE – V SEMESTER

COURSE NAME: 53S ELECTIVE III MACHINE LEARNING

LIST OF PRACTICALS

DATE : 28/10/2025 - FN **TIME : 3 HOURS**

1. **Perform Basic Functions using the College.csv dataset.**
   • Generating a numerical summary • Visualizing the pairwise relationships
   • Creating a new qualitative variable
   • Generating histograms with differing numbers of bins for select quantitative variables.

2. **EDA using Auto Dataset** by classifying predictors, calculating summary statistics (range, mean, standard deviation) for quantitative variables, manipulating the data by subsetting, and graphically investigating relationships between predictors, particularly focusing on their potential utility in predicting gas mileage (mpg).

3. **Perform Linear Regression using Boston Dataset:**

   • To investigate the relationship between per capita crime rate (crim) and all other variables in the Boston dataset
   • Fitting simple linear regression models for each predictor individually to determine statistically significant associations.
   • Fitting a multiple linear regression model using all predictors to identify which variables have a statistically significant non-zero coefficient (reject $H0:\beta j=0$) in the presence of other predictors.

4. **Perform Multiple Linear Regression using Auto Dataset** by gas mileage (mpg) as the response and all other variables (excluding name) as predictors. This includes visually inspecting pairwise relationships via a scatterplot matrix, quantifying linear associations with a correlation matrix, fitting the model, and diagnostically assessing the fit for violations of assumptions, outliers, and high leverage points.

5. **Perform Multiple Linear Regression using Carseats Dataset.**
   i. Fitting a multiple linear regression model.
   ii. Interpreting the meaning of the model coefficients, particularly those associated with the qualitative variables.
   iii. Writing the model in its correct mathematical equation form.
   iv. Determining which predictors have a statistically significant association with Sales.
   v. Refitting a reduced model using only the statistically significant predictors.

6. **Logistic Regression using Auto Dataset**.
    i. Creating the binary response variable mpg01.
    ii. Graphically identifying the most relevant predictors for mpg01.
    iii. Splitting the data into training and test sets.
    iv. Fitting a logistic regression model to the training data and evaluating its test error rate.

7. **Classification using Boston Dataset.**
    Predict whether a given suburb's crime rate (crim) is above or below the median. The analysis will involve fitting models with various subsets of predictors to identify which combination yields the best predictive performance.

Combinations:

{(1,5) ,(1,2),(3,4),(5,6),(2,4),(6,7)}

Internal Examiner Signature

External Examiner Signature